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United States Patent
Kind Code
Date of Patent
Inventor(s)

12385861
B2
August 12, 2025
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Charged particle detector, charged particle ray device, radiation detector, and radiation detection device

Abstract

Provided are a charged particle detector and a radiation detector capable of obtaining an observation image with correct contrast without saturation even when the number of signal electrons incident on a detector is increased due to an increase in the current of a primary electron beam. The charged particle detector is characterized by having a scintillator (109) having a signal electron detection surface (109a) for detecting signal electrons emitted when a specimen is irradiated with primary electrons and converting the signal electrons into light, a light detector (111) having a light detection surface (111a) for detecting the light emitted from the scintillator (109), and a light guide (110) disposed between the scintillator (109) and the light detector (111), wherein the area of the light detection surface (111a) is larger than the area of the signal electron detection surface (109a).

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Appl. No.:	17/791933
Filed (or PCT Filed):	March 02, 2020
PCT No.:	PCT/JP2020/008668
PCT Pub. No.:	WO2021/176513
PCT Pub. Date:	September 10, 2021

Prior Publication Data

Document Identifier

US 20230050424 A1

Publication Date

Feb. 16, 2023

Publication Classification

Int. Cl.: G01N23/2251 (20180101); G01T1/20 (20060101)

U.S. Cl.:

CPC G01N23/2251 (20130101); G01T1/2002 (20130101); G01N2223/07 (20130101); G01N2223/40 (20130101); G01N2223/505 (20130101)

Field of Classification Search

CPC: G01N (23/2251); G01N (2223/07); G01N (2223/40); G01N (2223/505); G01T (1/2002); H01J (2237/2443); H01J (2237/24465); H01J (37/244)

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a charged particle detector, a charged particle ray device, a radiation detector, and a radiation detection device.

BACKGROUND ART

(2) Detectors are used to convert particle signals such as electrons and ions and radiation signals such as X-rays and gamma rays into voltage, current signals, and the like. When the detection target is a charged particle, it is called a charged particle detector, and when the detection target is a radiation, it is called a radiation detector. For example, in a charged particle device using a charged particle ray such as an electron beam of a scanning electron microscope (SEM), and the like, a signal to be detected is a charged particle such as an electron, and a charged particle detector is essential. In the SEM, a specimen to be observed is irradiated with an electron beam generated by an electron source, whereby electrons emitted from the specimen are detected by the charged particle detector. The charged particle detector outputs a current according to the amount of detected electrons. An SEM image is formed by two-dimensionally displaying the relationship between the current amount and the electron beam irradiation position on the specimen.

(3) Many of such charged particle detectors are configured by a scintillator that converts the detected electrons into photons, a photodetector that detects the photons from the scintillator and converts the photons into current, and a light guide that delivers light emitted by the scintillator to the photodetector. As the photodetector, a photomultiplier tube (PMT), a multi-pixel photon counter (MPPC), or the like is used. In the radiation detector, only the type of scintillator is different in a similar configuration. That is, the radiation detector uses a scintillator that converts the detected radiation into light having a wavelength detectable by the photodetector, and is configured to deliver light from the scintillator to the photodetector via the light guide.

(4) In order to acquire more signals, it is required to arrange a detector in the vicinity of the specimen. For example, in the SEM, it is conceivable to arrange a detector between a specimen and an objective or a final lens. PTL 1 proposes such a detector.

CITATION LIST

Patent Literature

(5) PTL 1: JP 2018-152232 A

SUMMARY OF INVENTION

Technical Problem

(6) In the SEM, when the current of the primary electron beam is increased, the number of electrons exited from the specimen and detected increases, the S/N of the image improves, the noise of the image becomes inconspicuous, and the resolution enhances, and hence there is a

demand for increasing the current of the primary electron beam. At this time, the number of photons converted from electrons to photons by the scintillator and enters the photodetector also increases. However, when the incident photon density with respect to the area of the detection surface increases, the photodetector saturates and cannot accurately count the number of incident photons. In particular, in MPPC (e.g., manufactured by Hamamatsu Photonics K.K., model: S13360-6025C5), fine quadrangular detection pixels having a side of about several tens of μm are spread on a quadrangular detection surface having a side of about several mm, and when a photon enters each detection pixel, an electric signal is generated for each pixel, where the electric signal of each pixel represents one detection of the photon. However, when the incident photon density increases and a plurality of photons simultaneously enter the detection pixel, the proportional relationship between the number of incident photons and the output current collapses, and an accurate photographed image cannot be obtained. That is, there is a problem that the photodetector saturates when the current of the primary electron beam increases.

(7) PTL 1 described above does not describe an increase in current of a primary electron beam, and does not describe a problem related to saturation. The problem that the signal saturates is not limited to the charged particle detector, and the same applies to the radiation detector.

(8) In view of the above circumstances, an object of the present invention is to provide a charged particle detector and a radiation detector capable of obtaining an observation image with correct contrast without saturating even when the number of signal electrons entering the detector is increased due to an increase in the current of the primary electron beam.

Solution to Problem

(9) In order to achieve the above object, one aspect of the present invention provides a detector including a scintillator, a light guide, and a photodetector, in which an area of a detection surface of the photodetector is larger than an area of a detection surface of the scintillator.

(10) A more specific configuration of the present invention is described in the claims.

Advantageous Effects of Invention

(11) In view of the above circumstances, it is possible to provide a charged particle detector and a radiation detector capable of obtaining an observation image with correct contrast without saturating even when the number of signal electrons entering the detector is increased due to an increase in the current of the primary electron beam.

(12) Problems, configurations, and effects other than those described above will be made apparent by the description of the following embodiments.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a schematic diagram of a charged particle ray device (SEM) according to a first example.

(2) FIG. 2 is a schematic diagram of the charged particle detector of FIG. 1.

(3) FIG. 3A is a schematic diagram of the charged particle detector according to a second example.

(4) FIG. 3B is a schematic view of a detection surface of a photodetector viewed from a specimen direction of the charged particle detector in FIG. 3A.

(5) FIG. 4A is a schematic diagram of the charged particle detector according to a third example.

(6) FIG. 4B is a schematic view showing an example of a detection surface of a photodetector viewed from a specimen direction of the charged particle detector in FIG. 4A.

(7) FIG. 4C is a schematic view showing another example of a detection surface of a photodetector viewed from a specimen direction of the charged particle detector of the third example.

(8) FIG. 4D is a schematic view for explaining a first example of a divided structure of the scintillator and a method for manufacturing the same.

- (9) FIG. 4E is a schematic view showing a second example of the divided structure of the scintillator.
- (10) FIG. 4F is a schematic view for explaining a method for manufacturing the scintillator of FIG. 4E.
- (11) FIG. 4G is a schematic view showing a third example of the divided structure of the scintillator.
- (12) FIG. 5 is a schematic diagram of the charged particle detector according to a fourth example.
- (13) FIG. 6 is a schematic diagram of the charged particle detector according to a fifth example.
- (14) FIG. 7 is a schematic diagram of the charged particle detector according to a sixth example.
- (15) FIG. 8 is a schematic diagram of the charged particle detector according to a seventh example.
- (16) FIG. 9 is a schematic diagram of the charged particle detector according to an eighth example.
- (17) FIG. 10 is a schematic view showing an example of a positional relationship between a divided cell of a scintillator and a specimen.
- (18) FIG. 11 is a schematic view showing another example of a positional relationship between a divided cell of a scintillator and a specimen.
- (19) FIG. 12 is a schematic diagram of the charged particle detector according to a ninth example.
- (20) FIG. 13 is a schematic diagram of a charged particle ray device according to a tenth example.
- (21) FIG. 14 is a schematic diagram of a charged particle device according to an eleventh example.
- (22) FIG. 15 is a schematic diagram of an example of a radiation detection device of a twelfth example.
- (23) FIG. 16 is a diagram showing the radiation detector of FIG. 15.

DESCRIPTION OF EMBODIMENTS

(24) Hereinafter, examples of the present invention will be described in detail with reference to the drawings. In all the drawings for describing the examples, the same elements are denoted by the same reference numerals, and repeated description thereof will be omitted. Hereinafter, an example in the detection of the electron beam will be described, but the effect of the present invention is not lost even when ion beam or radiation is used.

First Example

- (25) FIG. 1 is a schematic diagram of a charged particle ray device (SEM) according to a first example. As illustrated in FIG. 1, a charged particle ray device **100a** includes a scanning deflector **103** and an objective lens **104** arranged on a trajectory of primary electrons **102** extracted from an electron source **101**. A specimen **106** arranged on a specimen transport stage **105** is irradiated with the primary electrons **102**, and a signal electron **107** is exited from the specimen **106**. Here, the signal electron **107** refers to an electron exited from the specimen, such as a secondary electron directly excited by the primary electrons and emitted into vacuum, or a reflected electron obtained in a state that the primary electrons are repeatedly scattered in the specimen and emitted into the vacuum again.
- (26) A charged particle detector **108** that detects signal electrons is provided below the objective lens **104**, and an opening **118** is provided at the center thereof to allow the primary electrons **102** to pass therethrough. The primary electrons **102** emitted from the electron source **101** are controlled by the objective lens **104** and focused on the specimen **106** so that the beam diameter is minimized. The scanning deflector **103** is controlled by a system control unit **120** so that the primary electrons **102** scan a defined region of the specimen **106**. The signal electrons **107** generated from the position where the primary electrons **102** have reached on the specimen **106** are detected by the charged particle detector **108**. In synchronization with a scanning signal sent from the system control unit **120** to the scanning deflector **103**, signal processing of the detected signal electrons **107** is performed, whereby an SEM image is formed on a monitor **121**.
- (27) FIG. 2 is a schematic diagram of the charged particle detector of FIG. 1. The charged particle detector of the present example includes a scintillator **109** that converts the incident signal electron **107** into light **112**, a light guide **110** that is provided in contact with the scintillator and guides the

light **112** to the photodetector, and a photodetector **111** that is provided immediately above the light guide and detects the guided light. Then, an area of a detection surface **111a** of the photodetector is larger than an area of a detection surface **109a** of the scintillator, and an optical path is expanded by the light guide **110**. The signal electrons **107** exit from the specimen **106** enter the scintillator **109**, but there is a bias in the incident position, and a large number of signal electrons enter a position close to the trajectory of the primary electrons **102**, that is, a position several mm from the opening **118**. On the other hand, in the photodetector, when the incident photon density with respect to the area of the detection surface increases, the proportional relationship between the incident photon amount and the output current collapses, the number of incident photons cannot be accurately counted, and an accurate photographed image cannot be obtained. When the current of the primary electron beam is increased, in particular, the number of photons entering from the detection surface **111a** close to the opening **118** increases, and the signal intensity corresponding to the primary electrons **102** entering near the opening **118** saturates.

(28) The charged particle detector **130** of the present example has a structure in which the optical path is expanded by the light guide, and thus has an effect that the photons obtained by converting signal electrons that have entered the scintillator at high density are dispersed, the density of the photons entering the photodetector is reduced, the photodetector does not saturate even when the current of the primary electron beam is increased and a photographed image with correct contrast can be obtained.

(29) In addition, in order to allow as many signal electrons **107** as possible to enter, it is desirable to make the opening **118** as small as possible within a range not interfering with the trajectory of the primary electrons **102**. Therefore, with respect to the detection surface **109a** of the scintillator, the detection surface **111a** of the photodetector is expanded toward the outer side without being expanded toward the opening **118** side. In the charged particle detector of the present example, the opening **118** is made as small as possible within a range not interfering with the trajectory of the primary electrons, and the detection surface of the photodetector is expanded toward the outer side with respect to the detection surface of the scintillator, so that an effect that a large amount of signal electrons **107** can be detected is obtained.

(30) Furthermore, the inventor has found that the transmittance of the light guide (the ratio of the light exiting from the other surface with respect to the light entering from the one surface) becomes higher when the light is propagated from the side having the smaller area to the side having the larger area. Therefore, when the area of the exit surface is increased with respect to the incident surface as in the present configuration, not only is the light dispersed but the transmittance is also increased.

(31) Note that the effect of the present invention can be obtained as long as the area of the light detection surface **111a** is larger than the area of the signal electron detection surface **109a**, and thus the shape of the light guide **110** is not limited. The light guide **110** of FIG. 2 has a shape in which the transverse section of the light guide **110** increases from the signal electron detection surface **109a** toward the light detection surface **111a**, but for example, may have a shape in which the transverse section decreases from the signal electron detection surface **109a** and increases from a certain location toward the signal electron detection surface **109a**.

Second Example

(32) FIG. 3A is a schematic diagram of a charged particle detector according to a second example, and FIG. 3B is a schematic view of a detection surface of a photodetector viewed from a specimen direction of the charged particle detector of FIG. 3A. In a charged particle detector **108b** of the present example, the photodetector **111** of the first example is an array-shaped photodetector **111b** including a plurality of detection cells **111c**, and the light guide **110** includes a plurality of divided blocks **110a**. As illustrated in FIG. 3B, in the present example, as an example, eight array-shaped photodetectors **111b** (e.g., manufactured by Hamamatsu Photonics K.K., model: S13615-1025N-04) including detection cells **111c** of 16 (4×4) channels are arranged to form a photodetector. In

addition, eight array-shaped photodetectors (e.g., manufactured by Hamamatsu Photonics K.K., model: S13615-1025N-08) including detection cells of 64 (8×8) channels can be arranged.

(33) The divided blocks **110a** of the light guide correspond to the detection cells **111c** of the array-shaped photodetector on a one-to-one basis, and the photon entering the divided block **110a** of one light guide cannot enter another adjacent block (strictly speaking, the photons slightly enter another adjacent block). With this structure, the signal electrons **107** that have entered the detection surface **109a** of the scintillator are converted into photons, the photons enter the divided block **110a** of the light guide immediately above the incident position of the scintillator, and the light is guided in the divided block **110a** of the light guide to enter the detection cell **111c** of the array-shaped photodetector **111b** corresponding to the relevant divided block **110a**.

(34) Assuming that the distance from the incident position of the primary electrons **102** on the specimen **106** to the incident position of the signal electrons **107** on the detection surface **109a** of the scintillator is w , the distance from the surface of the specimen **106** to the detection surface **109a** of the scintillator is h , and the exit angle of the signal electrons **107** from the specimen is α , when the photodetector **111b** detects the signal electrons **107**, w can be measured from the position of the detection cell **111c**. Furthermore, since h is a known value, the exit angle α of the signal electron from the specimen can be calculated from w and h . Since the direction in which the signal electrons are emitted varies depending on the material and shape of the specimen, information on the material and shape of the specimen can be obtained by detecting the exit angle of the signal electrons from the specimen. The detector capable of detecting the incident position of the signal electron in this manner is referred to as a position discrimination detector in the present specification. With the configuration described above, the charged particle detector of the present example has a structure in which the incident position of the signal electron on the detection surface can be discriminated with satisfactory accuracy and the exit angle of the signal electron from the specimen can be calculated, and has an effect in that information on the material and shape of the specimen can be obtained.

Third Example

(35) FIG. 4 is a schematic diagram of a charged particle detector according to a third example. Since the signal electrons enter the scintillator to be converted into light, but the light propagates inside the scintillator, in the charged particle detector of the second example, photons converted from the signal electrons may move inside the scintillator, and may not enter the divided block **110a** of the light guide immediately above the incident position of the signal electrons on the scintillator but may enter the divided block of another adjacent light guide. As in this phenomenon, a phenomenon in which the position where the signal electron entered does not correspond to the position of the photodetector that receives the photon is referred to as crosstalk in the present specification. The correspondence relationship between the incident position of the signal electron on the scintillator and the position of the detection cell of the photodetector collapses due to the crosstalk (the above-described w cannot be accurately measured), and the exit angle α of the signal electron from the specimen cannot be accurately calculated from the position of the detection cell of the photodetector when the signal electrons have entered, and the accuracy of the incident position discrimination lowers.

(36) FIG. 4B is a schematic view of one example of a detection surface of the photodetector viewed from a specimen direction of the charged particle detector of FIG. 4A, and FIG. 4C is a schematic view of another example of a detection surface of the photodetector viewed from a specimen direction of the charged particle detector of the third example. In the charged particle detector of the present example, in addition to the structure of the second example, the scintillator **109** is divided into divided cells **109b** of a plurality of scintillators by a dividing unit **113**. Here, division means a structure that reduces propagation of light, for example, a structure in which a notch is provided in the dividing unit of the scintillator, the dividing unit is completely cut, or a partition is provided thereat.

(37) An opening **118** for emitting primary electrons is provided at the center of the scintillator **109**. The divided cells **109b** of the scintillator, the divided blocks **110a** of the light guide, and the detection cells **111c** of the photodetector correspond on a one-to-one basis, and photons do not enter adjacent divided cells or divided blocks while moving from the scintillator to the light guide and from the light guide to the photodetector. The scintillator **109** has an annular shape in FIG. **4B** and a quadrangular shape in FIG. **4C**. As described above, the shapes of the scintillator, the light guide, and the photodetector are not limited as long as the divided cells of the scintillator, the divided blocks of the light guide, and the divided cells of the photodetector correspond on a one-to-one basis.

(38) In a case where the array structures of the scintillator **109** and the photodetector are different, the light guide compensates for the difference in the array structure to have a one-to-one correspondence. As the scintillator is divided, the movement of photons from the divided cell of the scintillator to which the signal electrons entered to the adjacent divided cell can be reduced, and crosstalk can be reduced, so that when the signal electrons enter, the exit angle α of the signal electrons from the specimen can be calculated more accurately from the position of the detection cell of the photodetector. The charged particle detector according to the present embodiment has an effect that crosstalk can be reduced and accuracy of incident position discrimination of signal electrons can be improved by dividing the scintillator.

(39) In addition, by forming a reflection layer in the dividing unit **113** of the scintillator, signal electrons enter a divided cell of a certain scintillator and are converted into photons, and jump out from a side surface of the incident divided cell to enter an adjacent divided cell thus preventing crosstalk from occurring, and the exit angle α of the signal electrons from the specimen can be calculated more accurately from the position of the detection cell of the photodetector.

(40) As the material of the reflection layer, a material having a high light reflectance is preferably used, and specifically, aluminum (Al) is desirable. Examples of a method of forming the reflection layer include a method of forming a thin film made of a material constituting the reflection layer on the surface of the dividing unit **113** by sputtering, vapor deposition, or the like, a method of sticking a metal plate made of a material constituting the reflection layer, and a method of applying a resin mixed with particles having high reflectance. The charged particle detector according to the present embodiment has an effect that crosstalk can be reduced and accuracy of incident position discrimination of signal electrons can be improved by forming the reflection layer on the dividing unit **113** of the scintillator.

(41) A divided structure of the scintillator and a method for manufacturing the same will be described. FIG. **4D** is a schematic view for explaining a first example of a divided structure of the scintillator and a method for manufacturing the same. In the structure of FIG. **4D**, the divided cells **109b** of the scintillator are molded so as to coincide with the pitch of the divided blocks **110a** of the light guide, and are adhered with an adhesive **115**. When the reflection layer is formed, the crosstalk can be reduced by forming the reflection layer on the side surface **109c** of the divided cell **109b** of the scintillator.

(42) FIG. **4E** is a second example of a divided structure of the scintillator, and FIG. **4F** is a schematic view for explaining a method for manufacturing the scintillator of FIG. **4E**. In the structure of FIG. **4E**, a partition frame **113a** is created to coincide with the pitch of the divided block of the light guide, and the divided cells **109b** of the scintillator are formed in the partition frame **113a** at a similar pitch. In the case of forming the reflection layer, the partition frame **113a** is made of a material having high reflectance such as Al. In the method for manufacturing of FIG. **4F(a)**, the divided cells **109b** of the scintillator molded at a similar pitch as that of the partition frame are fitted into and fixed to the partition frame **113a**, and are arranged such that the pitch of the divided cells of the scintillator and the pitch of the divided blocks of the light guide coincide with each other. In the method for manufacturing of FIG. **4F(b)**, the partition frame **113a** is fixed in advance so as to coincide with the pitch of the divided blocks of the light guide, and the divided

cells **109b** of the scintillator made of powder phosphor (e.g., P47 (Y.sub.2SiO.sub.5: Ce), YAG, GGAG ((Y,Gd).sub.3(Al,Ga).sub.5O.sub.12: Ce, (Y,Gd).sub.3(Al,Ga).sub.5O.sub.12: Tb), YAP (YAlO.sub.3: Ce), GOS(Gd.sub.2O.sub.2S: Pr,Gd.sub.2O.sub.2S: Ce, Gd.sub.2O.sub.2S: Tb)) are formed in the partition frame **113a**. As a method of forming the powder phosphor, a sedimentation applying method or a printing method can be considered.

(43) In the structure of FIG. **4E**, since the divided cells **109b** of the scintillator are integrated by the partition frame **113a**, the scintillator can be easily fixed without adhering to the light guide. The charged particle detector of the present example has a structure in which the divided cells of the scintillator are integrated and fixed by the partition frame without using an adhesive, so that the manufacturing process can be simplified and the manufacturing cost can be reduced, and there is no risk that the inside of the vacuum chamber in which the scintillator is disposed is contaminated by gas from the adhesive.

(44) FIG. **4G** is a schematic view showing a third example of the divided structure of the scintillator. In the structure of FIG. **4G**, first, grooves **117** are provided in the scintillator **109** so as to coincide with the pitch of the divided blocks of the light guide, a material (resin etc.) having an adhesive force is impregnated into the grooves to form the partition **113b**, and the lower portion of the scintillator **109** is removed until the partition **113b** is reached thereby forming the divided cells **109b** of the scintillator. In the case of forming the reflection layer, a resin or an adhesive obtained by mixing particles having high reflectance to a material to be the partition **113b** is used.

(45) In the method of FIG. **4G** as well, since the divided cells **109b** of the scintillator are integrated, the scintillator can be fixed without adhering to the light guide, similar to the method of FIG. **4F**. The charged particle detector of the present example has the effect that the manufacturing process can be simplified and the manufacturing cost can be reduced by integrating and fixing the divided cells of the scintillator with resin or adhesive to be the partition between the divided cells of the scintillator.

(46) In addition, since the pitch of one detection cell of the existing array-shaped photodetector is about 1 mm at the minimum, the resolution of position discrimination of the charged particle detector becomes about 1 mm at the maximum when the areas of the detection cell of the photodetector and the divided cell of the scintillator are designed to be the same size. On the other hand, in the charged particle detector of the present example, by expanding the optical path with the light guide, the area of the divided cell of the scintillator can be made smaller than that of the detection cell of the photodetector, and the resolution of incident position discrimination of the signal electrons can be improved as the area of the divided cell of the scintillator is made smaller. The charged particle detector of the present example has an effect that the resolution of incident position discrimination of the signal electrons can be improved by expanding the optical path with the light guide, and having the area of the divided cell of the scintillator smaller than that of the detection cell of the photodetector.

Fourth Example

(47) FIG. **5** is a schematic diagram of a charged particle detector according to a fourth example. The charged particle detector of the present example also has a structure in which crosstalk is reduced by dividing the scintillator **109** similarly to the second example. In the charged particle detector of the third example, the divided cells of the adjacent scintillators are completely separated, but in the charged particle detector of the present example, the scintillator is divided by providing the groove **117** in the scintillator. As the scintillator is divided by the groove, the crosstalk can be reduced, and the exit angle α of the signal electrons from the specimen can be more accurately calculated from the position of the detection cell of the photodetector when the signal electrons enter.

(48) As the depth of the groove becomes deeper, crosstalk can be prevented, but the risk of the scintillator separating increases. Among the scintillators, there is a scintillator (e.g., a GaN scintillator) in which a light emitting layer is formed on a base material. In the GaN scintillator, a

light emitting layer called a multi-quantum well (MQW) layer is formed on a base material such as sapphire. In the case of using a scintillator in which a light emitting layer is formed on such a base material, it is desirable to provide a groove so that the light emitting layer is divided. In addition, since the divided cells **109b** of the scintillator are integrated, the light guide and the scintillator can be easily aligned and fixed by forming the grooves so as to coincide with the pitch of the divided blocks of the light guide in advance, and the manufacturing process can be simplified, and the manufacturing cost is reduced.

(49) The charged particle detector according to the present example has an effect that the manufacturing process can be simplified, the manufacturing cost can be reduced, and the accuracy of incident position discrimination of signal electrons can be improved according to a structure of dividing the scintillator with grooves.

(50) In addition, by forming a reflection layer in the groove **117**, signal electrons enter a divided cell of a certain scintillator and are converted into photons, and jump out from a side surface of the divided cell to which the signal electrons have entered to enter an adjacent divided cell thus preventing crosstalk from occurring.

(51) A material example of the reflection layer and a method of forming the same are similar to those of the third example. The charged particle detector according to the present example has an effect that crosstalk can be reduced and accuracy of incident position discrimination of signal electrons can be improved by forming the reflection layer in the groove **117** of the scintillator.

Fifth Example

(52) FIG. **6** is a schematic diagram of a charged particle detector according to a fifth example. The charged particle detector of the present example uses a tapered fiber optics plate (FOP) **114** as a light guide in addition to the structure of the charged particle detector of the first example. The FOP is an optical component in which fine optical fibers having a diameter of several μm are bundled (e.g., manufactured by Hamamatsu Photonics K.K., model: J5734). Since the photons that have entered the fiber of the incident surface of the FOP exit from the same fiber of the exit surface without invading the adjacent fiber, the photons can be transmitted to the exit surface while holding the incident position of the photons on the incident surface.

(53) In the charged particle detector of the present example, the FOP has a tapered shape, and the area of the exit surface and the fiber diameter are larger than those of the incident surface, but the fact that photons can be transmitted while holding the incident position information of the photons on the incident surface is the same. As described above, there is a bias in the incident position of photons on the scintillator, and most of the signal electrons enter a position close to the trajectory of the primary electrons, that is, a position several mm from the opening **118**, but the charged particle detector of the present example can guide the photons that have entered with high density to the photodetector while dispersing the photons by fine optical fibers of the tapered FOP, and can prevent the photodetector from being saturated. As described above, the charged particle detector of the present example has a structure in which the optical path is expanded by the tapered FOP, and thus has an effect that the photodetector does not saturate even when the current of the primary electron beam is increased and a photographed image with correct contrast can be obtained.

(54) Furthermore, the transmittance of the tapered FOP (the ratio of the light exiting from the other surface with respect to the light entering from the one surface) becomes higher when the light is propagated from the side having the smaller area to the side having the larger area. Therefore, when the area of the exit surface is increased with respect to the incident surface as in the present configuration, not only is the light dispersed but the transmittance is also increased.

Sixth Example

(55) FIG. **7** is a schematic diagram of a charged particle detector according to a sixth example. In the charged particle detector of the present example, an array-shaped photodetector and a divided scintillator are used in addition to using the tapered FOP **114** for the light guide similarly to the fourth example. Similar to the charged particle detectors of the third example and the fourth

example, the charged particle detector of this example has the divided cells **109b** of the scintillator and the detection cells **111c** of the photodetector corresponding to each other on a one-to-one basis, and has a function as a position discrimination detector capable of detecting the exit angle of the signal electrons from the specimen.

(56) In the case of using a light guide including a plurality of divided blocks as in the charged particle detectors of the third example and the fourth example, it is necessary to align all of the divided cells of the scintillator, the divided blocks of the light guide, and the detection cells of the photodetector with high accuracy. On the other hand, when the fiber diameter of the tapered FOP is several μm and the fiber diameter of the FOP is sufficiently small compared to the size of the divided cell of the scintillator or the detection cell of the photodetector, the divided cells of the scintillator and the detection cells of the photodetector are respectively aligned with high accuracy, and when the magnification (area ratio between the incident surface and the exit surface) of the tapered FOP is equal to the area ratio between the detection surface **109a** of the scintillator and the detection surface **111a** of the photodetector, the tapered FOP does not need to be aligned with high accuracy, and the assembly process and structure of the detector can be simplified.

(57) The charged particle detector of the present example has a structure in which the tapered FOP is used for the light guide, so that the light guide does not need to be aligned, the manufacturing process of the detector can be simplified, and the manufacturing cost can be reduced.

Seventh Example

(58) FIG. **8** is a schematic diagram of a charged particle detector according to a seventh example. The charged particle detector of the present example has a structure in which the scintillator **109** and the detection surface of the photodetector **111** are arranged at positions (perpendicular in the drawing) that are not parallel to each other and light is guided by the light guide **110** in the configuration of the first example. The charged particle detector needs to be as small as possible so that it can be installed between the specimen and the pole piece of the final lens as described above. The charged particle detector of the present embodiment has an effect that it is not always necessary to arrange the scintillator **109** and the photodetector **111** in parallel by bending the light guide, and the degree of freedom of the shape of the charged particle detector is increased, and an effect that the charged particle detector can be accommodated in a small space between the objective lens and the specimen.

Eighth Example

(59) FIG. **9** is a schematic diagram of a charged particle detector according to an eighth example. The charged particle detector of the present example is obtained by forming the charged particle detector of the third example in a dome shape. The scintillator **109** is divided by the dividing unit **113**, and the area of the detection cell **111c** of the photodetector is larger than that of the divided cell **109b** of the scintillator.

(60) FIG. **10** is a schematic diagram showing an example of a positional relationship between a divided cell of a scintillator and a specimen. In FIG. **10**, the divided cells of the scintillator are arranged parallel to the specimen. When the divided cells of the scintillator are arranged parallel to the specimen as in the second to fifth examples, the positional relationship between the divided cells of the scintillator and the specimen is as shown in FIG. **10**. Assume that a solid angle drawn by the signal electron entering the divided cell **111c1** of the scintillator closest to the trajectory of the primary electrons **102** is ω_1 , and a solid angle drawn by the signal electron entering the divided cell **111c2** of the farthest scintillator is ω_2 . In the arrangement as illustrated in FIG. **10**, as the divided cell of the scintillator is farther from the trajectory of the primary electrons **102**, the distance from the incident position of the primary electrons on the specimen to the divided cell of the scintillator becomes longer, and the solid angle also becomes smaller. Therefore, ω_2 is smaller than ω_1 , and the number of signal electrons entering the divided cell **111c2** of the scintillator farthest from the trajectory of the primary electrons **102** decreases.

(61) FIG. **11** is a schematic view showing another example of a positional relationship between a

divided cell of a scintillator and a specimen. In FIG. 11, the divided cells of the scintillator are arranged parallel to the specimen. When the scintillator is arranged in a dome shape with respect to the specimen as in the present example, the positional relationship between the divided cells of the scintillator and the specimen is as shown in FIG. 11. In the arrangement as illustrated in FIG. 11, even when the distance from the trajectory of the primary electrons 102 to the divided cell of the scintillator changes, the distance from the incident position of the primary electrons on the specimen to the divided cell of the scintillator becomes equal, and the solid angle also becomes equal. Therefore, ω_2 is equal to ω_1 , and the number of signal electrons entering the divided cell 111c2 of the scintillator farthest from the trajectory of the primary electrons 102 increases compared to the case of FIG. 10. The charged particle detector of the present example has a structure in which the divided cells of the scintillator are arranged in a dome shape, and thus has an effect that the number of signal electrons entering the divided cells of the scintillator far from the trajectory of the primary electrons increases.

Ninth Example

(62) FIG. 12 is a schematic diagram of a charged particle detector according to a ninth example. The charged particle detector of the present example includes a scintillator 109, a light guide 110, and a photodetector 111, and a plurality of charged particle detectors 108a in which an area of a detection surface 111a of the photodetector is larger than that of a detection surface 109a of the scintillator are arranged in a dome shape at positions where signal electrons 107 come flying. The charged particle detector of the present example has a function as a position discrimination detector, and since the charged particle detectors 108a are completely independent from each other, there is no crosstalk between the charged particle detectors, and the exit angle of the signal electrons 107 from the specimen can be more accurately detected from the position of the charged particle detector when the signal electrons enter.

(63) In addition, similarly to the eighth example, the distances from the incident positions of the primary electrons on the specimen to the scintillators of the respective charged particle detectors become equal, the solid angles drawn by the signal electrons entering the scintillators of the respective charged particle detectors also become equal, and the number of signal electrons entering the charged particle detector farthest from the trajectory of the primary electrons 102 increases. The charged particle detector of the present example has a structure in which a plurality of charged particle detectors in which an area of the detection surface of the photodetector is larger than that of the detection surface of the scintillator are arranged in a dome shape, and thus has an effect that the accuracy of the incident position discrimination of the signal electrons improves and the number of signal electrons entering the scintillator of the charged particle detector far from the trajectory of the primary electrons increases.

Tenth Example

(64) FIG. 13 is a schematic diagram of a charged particle ray device according to a tenth example. A scanning deflector 103 and an objective lens 104 are provided on the trajectory of the primary electrons 102 extracted from the electron source 101. The specimen 106 arranged on the specimen transport stage 105 is irradiated with the primary electrons 102, and signal electrons 107a having a small reflection depth at the specimen and signal electrons 107b having a large reflection depth at the specimen are exited from the specimen 106.

(65) An E×B deflector 116 is arranged on the objective lens 104 to deflect the signal electrons according to the magnitude of energy. The charged particle detector 108 having a function as a position discrimination detector including the divided scintillator 109, the divided light guide 110, the array-shaped photodetector 111b, and the dividing unit 113 is provided in the advancing direction of the deflected reflected electrons, and signal processing of the detected signal electrons is performed in synchronization with a scanning signal sent from a system control unit 1001 to the scanning deflector 103, so that an observation image is formed on a monitor 121.

(66) The E×B deflector 116 is a deflector using the behavior of electrons in an electric field and a

magnetic field, and has a function of not deflecting electrons (primary electrons) entering from above but deflecting electrons (signal electrons) entering from below. Among the signal electrons, there are reflected electrons that are exited from the specimen after the primary electrons are reflected in the specimen. Reflected electrons generally refer to electrons having energy of greater than or equal to 50 eV, and the magnitude of the energy varies depending on the reflection depth in the specimen. The energy of the reflected electrons decreases as the reflection depth in the specimen increases, and increases as the reflection depth decreases. The angle deflected by the $E \times B$ deflector varies depending on the magnitude of the energy of the electrons, and the signal electrons **107a** having a small reflection depth at the specimen have a large energy and are deflected at a small angle, and the signal electrons **107b** having a large reflection depth at the specimen have a small energy and are deflected at a large angle.

(67) Similarly to the position discrimination detector (e.g., third, fourth, sixth, and eighth examples), the charged particle detector **108** of the charged particle ray device of the present example can discriminate signal electrons from the position of the detection cell of the photodetector corresponding to the incident position of the signal electrons on the detection surface of the scintillator. In addition, the position discrimination detector can calculate the exit angle of the signal electrons from the specimen from the position of the detection cell of the photodetector when the signal electrons enter, but the charged particle detector of the charged particle device of the present example can similarly calculate the deflection angle of the signal electrons by the $E \times B$ deflector **116** from the position of the detection cell of the photodetector. Since the deflection angle by the $E \times B$ deflector **116** depends on the magnitude of the energy of the signal electrons, that is, the reflection depth, the reflection depth of the signal electrons can be calculated from the deflection angle by the $E \times B$ deflector **116**.

(68) As a result, the charged particle detector of the charged particle device of the present example can detect the signal electrons for each reflection depth from the position of the detection cell of the photodetector when the signal electrons enter, and can calculate the reflection depth. Furthermore, by forming an observation image using only the signal electrons having the same reflection depth, an observation image at a certain depth of the specimen can be acquired, and a three-dimensional observation image of the specimen can be acquired by acquiring the observation images at the respective depths and stacking the observation images in descending order of the reflection depth. The charged particle device of the present example has a structure in which the $E \times B$ deflector is arranged on the objective lens and the position discrimination detector is arranged in the advancing direction of the signal electrons deflected by the $E \times B$ deflector, and thus has an effect that the reflection depth of the signal electrons can be calculated, an observation image at a certain depth of the specimen can be acquired, and a three-dimensional observation image of the specimen can be acquired.

Eleventh Example

(69) FIG. **14** is a schematic diagram of a charged particle device according to an eleventh example. The charged particle ray device of the present example has a structure similar to that of the charged particle device of the tenth example, but is different in that a plurality of charged particle detectors **108a** in which an area of the detection surface **111a** of the photodetector is larger than that of the detection surface **109a** of the scintillator are arranged in a dome shape in the advancing direction of the signal electrons deflected by the $E \times B$ deflector **116**. With this structure, similarly to the charged particle device of the seventh example, the charged particle device of the present example can calculate the reflection depth of the signal electron from the position of the detection cell of the photodetector when the signal electrons enter, can acquire the observation image at a certain depth of the specimen, and can acquire the three-dimensional observation image of the specimen.

(70) Furthermore, by arranging the charged particle detectors **108a** in a dome shape, even if the distance from the trajectory of the primary electrons **102** to the charged particle detector **108a** changes, the distance from the deflection position of the signal electrons of the $E \times B$ deflector **116**

to the scintillator **109** of the charged particle detector **108a** becomes equal, and the number of signal electrons entering the scintillator **109** of the charged particle detector **108a** farthest from the trajectory of the primary electrons **102** increases. The charged particle detector of the present example has a structure in which the charged particle detectors are arranged in a dome shape, and thus has an effect that the number of signal electrons entering the scintillator of the charged particle detector far from the trajectory of the primary electrons increases.

(71) In addition, since the charged particle detectors **108a** of the charged particle device of the present example are completely independent from each other, there is no crosstalk between the charged particle detectors, the deflection angle by the $E \times B$ deflector **116** can be calculated more accurately, and the detection accuracy of the reflection depth of the signal electrons can be improved. The charged particle device of the present example has a structure in which a plurality of charged particle detectors in which an area of the detection surface of the photodetector is larger than that of the detection surface of the scintillator are arranged in a dome shape in the advancing direction of the signal electrons deflected by the $E \times B$ deflector, and thus has an effect that the detection accuracy of the reflection depth of the signal electrons improve.

Twelfth Example

(72) In the first to twelfth examples, the charged particle ray detector and the charged particle ray device have been described, but the configuration of the charged particle detector of the present invention can also be applied to a radiation detector. FIG. **15** is a schematic diagram of an example of a radiation detection device of the twelfth example, and FIG. **16** is a diagram illustrating the radiation detector of FIG. **15**.

(73) As illustrated in FIG. **15**, a radiation detection device **200** includes an X-ray source **201** that irradiates the specimen **106** on the specimen transport stage **105** with X-rays (radiation), and an X-ray detector **203** that detects the X-rays transmitted through the specimen **106**.

(74) As illustrated in FIG. **16**, the radiation detector includes a scintillator having a radiation detection surface for detecting radiation and converting the detected radiation into light, a photodetector having a light detection surface for detecting light emitted from the scintillator, and a light guide provided between the scintillator and the photodetector, and has a configuration in which an area of the light detection surface is larger than an area of the radiation detection surface. With such a configuration, similarly to the charged particle detectors of the first to eleventh examples described above, a radiation detector and a radiation detection device capable of obtaining an observation image with correct contrast without saturating even when the intensity of radiation entering the detector is increased can be obtained.

(75) As described above, according to the present invention, it has been shown that the current of the primary electron beam can be increased.

(76) The above-described examples have been described in detail for the sake of easy understanding of the present invention, and are not necessarily limited to those having all the described configurations. Furthermore, a part of the configuration of a certain example can be replaced with the configuration of another example, and the configuration of a certain example can be added with the configuration of another example. In addition, it is possible to add, delete, and replace other configurations for a part of the configuration of each example.

REFERENCE SIGNS LIST

(77) **100a**, **100b**, **100c** charged particle ray device **101** electron source **102** primary electron **103** scanning deflector **104** objective lens **105** specimen transport stage **106** specimen **107** signal electron **107a** signal electron having small reflection depth **107b** signal electron having large reflection depth **108a**, **108b**, **108c**, **108d**, **108e**, **108f**, **108g**, **108h**, **108i** charged particle detector **108a** charged particle detector **109** scintillator **109a** detection surface of scintillator **109b** divided cell of scintillator **109c** side surface of divided cell of scintillator **110** light guide **110a** divided block of light guide **111** photodetector **111a** detection surface of photodetector **111b** array-shaped photodetector **111c** detection cell of photodetector **112** light **113** dividing unit **113a** partition frame

113b partition **114** tapered FOP **115** adhesive **116** E×B deflector **117** groove **118** opening **120** system control unit **121** monitor **130** charged particle detector w distance from incident position of primary electron on specimen to incident position of signal electron on detection surface of scintillator h distance from surface of specimen to detection surface of scintillator α exit angle of signal electron from specimen ω_1 solid angle drawn by signal electron entering divided cell of scintillator closest to trajectory of primary electron ω_2 solid angle drawn by signal electron entering divided cell of scintillator farthest from trajectory of primary electron **200** radiation detection device **201** X-ray source **202** X-ray **203** X-ray detector

Claims

1. A charged particle detector comprising: a scintillator having a signal electron detection surface for detecting signal electrons that exit from a specimen when the specimen is irradiated with primary electrons, and converting the signal electrons into light; a photodetector having a light detection surface for detecting light emitted from the scintillator; and a light guide serving as an optical path of the light, the light guide being provided between the scintillator and the photodetector, wherein an area of the light detection surface is larger than an area of the signal electron detection surface, wherein the light guide and the photodetector are divided into a plurality of cells, and wherein, in each of the divided cells of the light guide, the optical path is expanded from an incident surface of the light provided at an end portion on the scintillator side toward an exit surface of the light provided at an end portion on the photodetector side.
2. The charged particle detector according to claim 1, wherein the light guide and the photodetector are divided into a plurality of cells by a dividing unit.
3. The charged particle detector according to claim 1, wherein the scintillator is divided into a plurality of cells.
4. The charged particle detector according to claim 3, wherein the scintillator is divided into a plurality of cells by the dividing unit.
5. The charged particle detector according to claim 2, wherein a reflection layer that reflects light is provided on a surface of the dividing unit.
6. The charged particle detector according to claim 3, wherein the scintillator includes a plurality of grooves provided toward the photodetector.
7. The charged particle detector according to claim 6, wherein a reflection layer that reflects light is provided on a surface of the groove.
8. The charged particle detector according to claim 5, wherein a material of the reflection layer is aluminum.
9. The charged particle detector according to claim 1, wherein the light guide includes a plurality of fiber optic plates.
10. The charged particle detector according to claim 1, wherein an area ratio between an incident surface connected to the scintillator of the light guide and an exit surface connected to the light detection surface is equal to an area ratio between the signal electron detection surface and the light detection surface.
11. The charged particle detector according to claim 1, wherein an area ratio between the signal electron detection surface and the light detection surface is 1:20 or less.
12. The charged particle detector according to claim 1, wherein cross-sectional shapes of the scintillator, the photodetector, and the light guide are fan shapes.
13. The charged particle detector according to claim 1, wherein the photodetector is arranged perpendicular to the scintillator.
14. A charged particle ray device comprising: the charged particle detector according to claim 1; a control unit that controls scanning of the primary electrons; and a display unit that processes the signal electrons and displays an image of the specimen.

15. A radiation detector comprising: a scintillator having a radiation detection surface for detecting radiation and converting a radiation signal into light; a photodetector having a light detection surface for detecting light emitted from the scintillator; and a light guide serving as an optical path of the light, the light guide being provided between the scintillator and the photodetector, wherein an area of the light detection surface is larger than an area of the radiation detection surface, wherein the light guide and the photodetector are divided into a plurality of cells, and wherein, in each of the divided cells of the light guide, the optical path is expanded from an incident surface of the light provided at an end portion on the scintillator side toward an exit surface of the light provided at an end portion on the photodetector side.

16. A radiation device comprising: the radiation detector according to claim 15; a radiation source that irradiates a specimen with radiation; and a display unit that processes a signal obtained from the radiation detector and displays an image of the specimen.
